

GREEN HOMES FOR A GREATER NEW JERSEY

New Jersey promotes affordable, green homes through its Green Homes Office.

by SUSAN BASS LEVIN,
DARREN S. PORT, AND
ROBERT E. WISNIEWSKI

At the New Jersey Department of Community Affairs (DCA), we are committed to investing in our communities for a better New Jersey, and we work to achieve that goal through a variety of innovative programs. To ensure a great quality of life for residents, we know we must provide jobs, housing, and economic growth, while at the same time preserving and protecting our natural resources. In short, we must incorporate both economic growth and environmental preservation.

In 1998, DCA launched the state's first sustainable-development pilot program to test the viability of building energy-efficient, environmentally friendly affordable housing. The program required all units in the pilot to meet the standards of the Public Service Electric and Gas (PSE&G) Energy-Efficient Home (EEH) 5-Star Program. This program was one of the precursors to what is now known as the national EPA/DOE Energy Star Program. In an effort to promote a market-transforming approach to energy- and resource-efficient design, construction, and methodologies, 367 units were designed and built. The success of the pilot resulted in the creation of DCA's Green Homes Office (GHO) two years later.

Capitalizing on the opportunity to change energy use habits and further explore energy efficiency through the lens of green building, the GHO developed the New Jersey Affordable Green Program (NJAG). The only statewide green affordable housing program in the country, NJAG has become a national



The exterior of Bellevue Court Homes conceals the development's leading-edge technologies behind the traditional 1896 brick facades.

model for green affordable housing and has inspired architects and construction officials across the state to go green.

NJAG collaborates with DCA's affordable housing subsidy program, Balanced Housing, to fund appropriate projects. In addition to the substantial subsidy that Balanced Housing puts into a project to write down the cost of development, up to \$7,500 per unit is made available to a developer incorporating green materials and technologies. As project designs come to fruition, this subsidy is allocated to pay for the incremental cost of upgrades from conventional materials to high-performance green features.

In the last five years, incorporating a blend of Balanced Housing and Green

subsidies, the GHO has worked to increase the use of innovative green materials, design, and high-performance building technologies in over 2,400 New Jersey Affordable Green/New Jersey Energy Star-certified affordable homeownership and rental units in the state.

In addition to the heavy emphasis on achieving energy efficiency through fundamental building science strategies, NJAG aggressively addresses indoor air quality (IAQ) issues. By requiring the integration of low-VOC materials; healthy and durable flooring, including hardwood, ceramic, bamboo, linoleum, recycled rubber, and cork; and advanced ventilation systems, New Jersey renters and homeowners are provided with



The exterior of Dr. King Plaza features highly durable insulated fiberglass frame windows.



Dr. King Plaza uses 2 inches of R-14.3 rigid polyisocyanurate foam insulation on the outside of the studs.

Table 1. Brick Church Commons Project Summary: Energy-Efficient & Green Features

Building Envelope

Exterior walls: R-14.3 exterior polyisocyanurate rigid insulation

Roof: R-28.6 exterior rigid insulation

Windows: fiberglass frame, low-e, argon filled

Air sealing measures: airtight drywall approach, insulated foam sealant, flexible membrane sealant at window rough openings

Mechanical Systems

Heating system: 90% efficient condensing gas furnaces

Air conditioning: 1.5 tons, 12-SEER

Programmable thermostats

Central hot water with high-efficiency gas-fired water heaters; hot water circulated in a continuous loop

Appliances/Lighting

Energy Star refrigerator and dishwasher

Indoor lighting: 1 Energy Star light fixture per room

Green/Recycled Materials Practices

Postindustrial recycled wood flooring

Wood base molding made from postindustrial wood

IAQ Measures

Wood flooring in kitchens, living rooms, dining rooms, and hallways

Ceramic tile in bathrooms

Granite tile in lobbies

Mechanical compartmentalized ventilation in apartments and public hallways

Water Conservation

Drought-resistant landscaping

Transportation

Accessible to train and buses

excellent IAQ. Through the use of dual-flush toilets and low-flow fixtures, water usage is also reduced.

A Success Story in East Orange

Brick Church Commons, located in East Orange, New Jersey, features three new energy-efficient buildings. These buildings provide 104 units of housing and 37,500 ft² of retail space in a once-vacant downtown site in East Orange. Developed by RPM Development Group, Brick Church Commons combines a mix of affordable and market rate apartments above retail businesses. The project is in a pedestrian-friendly location with local shopping and access to buses and trains. Residents have on-site parking, laundry facilities, and a children's play area. The project's namesake, a red-brick church, stands to the west; its nineteenth-century spire creates an attractive focal point for the neighborhood. Buildings are clad in block at the ground floor level and red brick above.

One building within Brick Church Commons, Dr. King Plaza, contains 4 one-bedroom, 84 two-bedroom, and 8 three-bedroom apartments. Affordable units comprise 80% of the building, while the remaining 20% of the units are rented at market rate. In addition to the energy-efficient and green features consistent throughout all three buildings (see Table 1), Dr. King Plaza invites and promotes unique community involvement. The lobby acts as a gallery for the paintings of local artists and features a mural with excerpts from Dr. Martin Luther King Jr.'s famous, "I Have a Dream" speech. An attractive tree-lined, brick-paved promenade encourages foot traffic past the grade level shops. The downtown site offers residents close proximity to trains, buses, and local shopping. Underground parking is provided for residents. On-site parking for the project's retail customers is broken up with attractive landscaping and site lighting. Dr.

King Plaza also has a children's outdoor play area.

All apartments in the complex have generously sized kitchens and living rooms. High ceilings and open floor plans allow daylight to wash in through the spaces. Of the 104 units, 34 are for low-income residents and 68 are market rate. The affordable units have one and a half baths, while the market rate units have two full baths and an added den.

Green and Energy Efficient in East Orange

All kitchens, living rooms, dens, and hallways in these three buildings feature wood flooring made of postindustrial recycled wood. Base molding is also made from postindustrial wood. IAQ in the buildings is not compromised by excessive pollutants, since the buildings make minimal use of carpet (which is used only in Brick Church's bedrooms), and use only water-based wood floor finish, which has lower volatile organic compound (VOC) content than oil-based sealants. Bathrooms and laundry rooms have ceramic tile floors, and the building lobbies have granite floors. To maintain a healthier indoor environment, each apartment is ventilated separately from its neighbor and has a continuous ventilation system.

Buildings 1 and 2 at Brick Church Commons are four-story wood frame structures with R-13 fiberglass insulation inside the wall cavities. Building 3, or Dr. King Plaza, is five stories and is framed in steel studs. To avoid conductive heat losses through the steel framing, Dr. King Plaza uses 2-inch R-14.3 rigid polyisocyanurate foam insulation on the outside of the studs. The roof is insulated with 4 inches of rigid foam above the roof deck. This provides an R-value of 28.6 (or an aged value of R-24, R-6 per inch) and keeps the inside of the roof deck warm, reducing the potential for condensation on that surface.

The windows in each building have insulated fiberglass frames and heat-saving low-e argon-filled glass. Window openings are wrapped with a flexible sealing membrane to reduce air infiltra-

tion and the potential for water intrusion. The buildings are extensively air sealed, with foam sealant at the roof, floors, outside walls, windows, and walls between apartments.

The top-floor ceiling of building 1 is insulated with R-30 fiberglass insulation. Each apartment has a sealed-combustion, 90% efficient condensing gas furnace and a 12-SEER 1.5-ton air conditioning unit. All ductwork is sealed with mastic for airtightness and is insulated with R-4.2 duct wrap insulation.

Each apartment's heating and cooling system is controlled with programmable thermostats. Each building has central water heating provided by high-capacity energy-efficient sealed-combustion gas water heaters. Hot water is run in a continuous loop throughout the buildings to provide hot water almost immediately. All kitchen appliances are Energy Star rated, and most light fixtures use CFLs. The laundry areas on each floor have horizontal-axis, high-efficiency washing machines. All of the Brick Church Commons buildings comply with the New Jersey Energy Star program.

Efficient Rehab in Trenton

Bellevue Court Homes, located in Trenton, is New Jersey's first urban, rehab, affordable, energy-efficient housing development. Part of the larger 22-unit Energy Star Bellevue Court Project, the homes at 233 to 235 Bellevue Court conceal leading-edge technologies behind their traditional brick facades.

Spearheaded by the city of Trenton and developed by Isles, Incorporated, and Tara Construction Management Corporation, these near zero-energy homes are projected by modeling to use over 70% less energy than a code-compliant home



The flooring in the East Orange projects is made of recycled or postindustrial wood.

and up to 40% less energy than their Energy Star neighbors. The two 1,700 ft² units, originally built in 1896, have three bedrooms and 2.5 baths, and incorporate sustainable and recycled materials throughout (see Table 2, p. 22).

Heating and cooling losses were reduced through the building envelope with highly insulated walls and careful air sealing. The inside faces of the existing masonry walls were covered with rigid insulation, followed by a 1-inch air space and 2 x 4 framing. This assembly was then filled with damp-spray cellulose, achieving a whole-wall R-value of 18.5. Attic insulation is R-40 blown-in cellulose, with R-40 rigid insulation on the attic access hatch. All openings in the thermal envelope were carefully sealed, and gypsum board was installed using airtight drywall. This created an air barrier at the finished wall surface. Air leaks that may have been missed during construction were revealed and sealed during blower door testing. Foil-faced foam insulation surrounds the rough opening of each window to reduce heat loss in an area that usually loses a lot of heat through thermal bridging. The windows themselves have fiberglass frames, which are well



Highly insulated walls and careful air sealing are used in Bellevue Court.

insulated and are much more durable than the typical window frames used in affordable housing. The insulated glass includes a heat-saving low-e layer and argon gas fill. Despite all of these measures, the actual results were not what

were expected (see “Bellevue Court Homes: Lessons Learned”).

To help provide exceptional air quality in these homes, mechanical ventilation is accomplished using a heat recovery ventilator (HRV). The exhaust vents from each bathroom and kitchen are ducted to the HRV, while fresh air from the HRV feeds into the return plenum of the furnace.

Custom-designed overhangs on the south wall of the microload homes block the high summer sun and allow for passive solar heating during the winter. The window glass was selected to allow solar heat to enter in winter. (It has a solar heat gain coefficient, or SHGC, of >0.6 .) As a passive cooling strategy, a skylight with a hand-held crank was placed over the central stair to allow warm inside air to move up and out of

the house in summer. This also brings daylight into the area.

The reduced heating loads are met with a 90% efficient sealed-combustion furnace. Air conditioning is provided by a 13-SEER 1.5-ton unit. The duct layout to deliver the heated and cooled air is specially designed to minimize friction losses so that furnace fan energy will be reduced.

Hot water is generated by a combination of a solar thermal system and a sealed-combustion gas water heater. The solar system on the roof preheats the water for the gas water heater. In addition, a GFX heat exchanger recovers waste heat from the shower's graywater. The combination of the solar thermal unit and the GFX is projected to reduce water heating energy use by over 70%.

A 2.5 kW PV array on the roof is projected to offset close to two-thirds of

Bellevue Court Homes: Lessons Learned

Performance: The Green Homes Office is currently monitoring fuel usage data and will have a full energy analysis available this summer. Looking strictly at the energy usage of the two solar homes, the building is performing less efficiently than modeled. A few observations reveal interesting lessons.

Contractor's View: The contractor noted that it was difficult to combat a "rule of thumb" mentality for the sizing of heating and cooling systems. One consultant stated that only by insisting on *Manual J* calculations did they avoid installing a cooling system twice the size necessary. The developer noted that having the air sealing done properly was a struggle. He commented that "infiltration is one of the least understood aspects of building science." He went on to say that when team members were on site to observe blower door tests, the importance of air sealing became very clear.

Post occupancy observations: A forced hot air system was utilized in the

two buildings, which have long ducted runs in the unconditioned basement. During the heating season the long duct runs cool down quickly between furnace on-cycles. At start-up of the next cycle the cooled air is pushed into the units on call for heat. The high-efficiency furnace does not have an adjustable time delay or heat exchanger temperature setting for the fan.

Thermostat settings have to be adjusted to obtain proper on-off differentials and homeowners need training on operation of the programmable thermostats.

Homeowners complained of air temperature differences from room to room, due to unbalanced air flows. In the course of correcting this problem, we discovered registers that were improperly installed and were also out of reach of the homeowner. Air flows should be balanced by the HVAC contractor, using a flow head to insure accurate measurement.

Homeowner behavior is a significant factor in energy consumption. Both owners are using electric heaters on the first floor because of the cool air from the furnace, both have 240 kWh/year freezers in the basement, and both have multiple televisions and other electronics with significant plug loads. One homeowner is utilizing an electric fireplace.

We observed that the homeowners always had shades drawn on the south facade which was designed for direct solar gain "sun tempering." We came to understand this was due to security and privacy reasons; security vs. sun in urban areas is a factor that should not be overlooked.

The PV system on one home is shaded by an adjacent home, creating less than optimum conditions for electric generation. Lesson: check solar access thoroughly.

Table 2. Bellevue Court Houses Project Summary: Energy-Efficient & Green Features

Building Envelope

Exterior walls: R-5 rigid insulation and damp-spray cellulose, whole-wall R-value of 18.5

Ceiling: R-40 blown-in cellulose

Windows: fiberglass frame, double-glazed, low-e, argon filled, SHGC>0.6

Exterior doors: insulated

Basement ceiling: insulated with R-19 encapsulated fiberglass

Air sealing measures: airtight drywall approach; nonexpanding foam at door and window rough openings; air sealing at basement ceiling

Mechanical Systems

Heating system: 90% efficient condensing gas furnace

Air conditioning: 1.5 tons; 13-SEER

High-efficiency sealed-combustion gas water heater

Solar domestic water heating: two 4 ft x 8 ft collectors with 119-gal storage tank

Heat recovery ventilator

Automatic setback thermostat

2.5 kW PV array

GFX drain water heat exchanger

Appliances/Lighting

Energy Star refrigerator and dishwasher

Indoor Lighting: 1 Energy Star light fixture with CFLs per room

Outdoor lighting: high-efficiency lighting with motion sensor

Green/Recycled Materials Practices

Recycling area provided in kitchen

Recycling storage area provided at rear or side of house

Job site recycling and reuse of materials during construction

Reuse of brick from site for pervious paving

Recycling of architectural elements that could not be reused on the project

Recycled-content plastic/wood lumber for porches and decks

PET carpet

ACQ instead of CCA pressure-treated lumber

IAQ Measures

Zero- and low-VOC paints

Track-off mat system at entry

Plywood kitchen cabinets

Low- and nontoxic wood finishes, caulks, and adhesives

Wood flooring in living rooms and dining rooms, ceramic tile in kitchens and baths

Sealed-combustion furnace and water heater

Ducted range hood

Ventilation to each occupied room, exhaust from moisture-producing rooms

Water Conservation

Dual-flush lower-water use toilets

Low-flow showerheads and faucets

Droughtproof plantings and grasses in species native to the area

Rainwater collection for gardening

Transportation

Bicycle storage in basement with grade level exit, or in shed



South wall overhangs provide cooling effects in the summer and passive solar heating in the winter.

the electric load in the house. The system will be net metered, allowing unneeded solar-produced electricity to be sold back to the utility.

Buyers of the microload houses will receive training on all the special features of their homes. The energy use and production of the NJGH houses will be monitored for a period of three years by the NJGH office to track the efficiency and performance of the units, as well as to provide annual inspections of the active and thermal solar units. The microload houses will be sold for \$55,000 to home buyers making 50% or less of the median income. Funding for the project was provided by the city of Trenton, the Balanced Housing program, HUD HOME Grants, the Federal Home Loan Banks of New York and Pittsburgh, and the NJGH Office.

Plans for the Future

Looking into the future, the Green Homes Office has two exciting initiatives on the drawing board. The first is the development of a microload project in the city of Camden, which is undertaking a number of initiatives to improve its economic vitality and provide high-quality affordable housing to residents. The Camden Microload Project aims to utilize high-performance design principles and passive and active solar strategies to construct 25 housing units, which will have a near net-zero use of grid provided electrical power. DCA, with the assistance of nationally recognized experts, will demonstrate that zero-

energy principles can apply to affordable urban residences within DCA price, quality, and code requirements. The resulting designs and constructed residences in Camden will be highly replicable models for urban housing throughout the coastal Mid-Atlantic climatic region.

The second program is the New Jersey High Performance Homes Plus. This program is designed to support the green efforts of market rate builders and promote the benefits of high-performance home building and renovation. High Performance Homes Plus will establish a state green building standard, advance whole-system, energy-efficient building practices among builders, and educate consumers about the advantages of the features in their homes. The program will coordinate with other national green building programs to address bioregional issues, and provide New Jersey builders and residents with a one-of-a-kind program tailored to the specific needs of the state.

The Department of Community Affairs is also currently collaborating with the New Jersey Housing and Mortgage

Finance Agency to promote and incorporate energy efficiency, IAQ, and other green building strategies into units that serve as homes for New Jersey's most mentally and socially challenged individuals. This green building emphasis on special-needs projects is a unique plan to create the healthiest and most durable housing units possible for the most vulnerable and underprivileged population.

Finally, the GHO is leading an effort to organize a Green Policy Working Group to promote sustainable development through state policy via the coordination of state agencies and programs, sharing of technical expertise and information, and collaboration on policy tasks that promote a green-built environment. The Policy Group will include state agencies and other stakeholders to develop statewide green policies for both affordable and market rate housing. Plans are also in the works to organize and sponsor a National Green Affordable Housing Conference right here in New Jersey.

Green buildings are cleaner, healthier, brighter, and more livable. They represent

the housing we need today to ensure a better tomorrow. Working together with local officials, architects, and builders, we are one step closer to preserving our future through the use of innovative green design and building technologies across the state. 🏠

Susan Bass Levin is commissioner of the New Jersey Department of Community Affairs and Darren S. Port is the director of the New Jersey Green Homes Office. Robert E. Wisniewski is a green building program specialist with the New Jersey Green Homes Office.

FOR MORE INFORMATION:

The New Jersey Green Homes Office
Tel: (609)292-3931
E-mail: njgreenhome@dca.state.nj.us
Web site: www.nj.gov/dca/dh/gho/index.shtml